

Figure 1: LEACE projection on a toy dataset in 3 steps. First the data is whitened using the linear transformation $W = (\Sigma_{XX})^{-1/2}$, ensuring equal variance in all directions. It is then orthogonally projected onto $\text{colsp}(W\Sigma_{XZ})^\perp$, guaranteeing linear guardedness. Finally, we unwhiten the data with the pseudoinverse W^+ so that its covariance structure mimics the original.

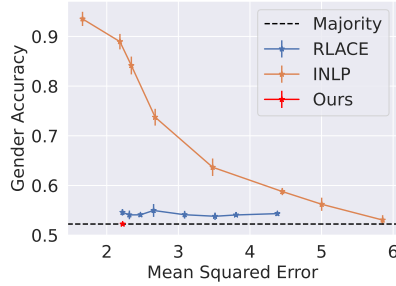


Figure 2: Gender prediction accuracy after debiasing against the mean squared distance of the debiased embedding from the original. Ours is the only method to achieve *perfect* erasure on this dataset (dashed line), and does so with low MSE. For INLP and RLACE, each point corresponds to a different setting of the rank hyperparameter; from left to right the ranks are 1, 4, 8, 16, 32, 50, 64, and 100. Our method is rank 1.

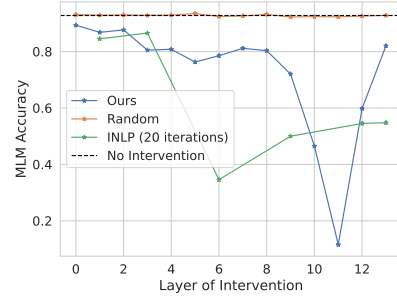


Figure 3: Corrected version of Figure 3 in our submission, which had erroneously reported that BERT exhibited *better* masked language modeling accuracy after a random subspace of its activations was ablated. One of our authors had accidentally transposed a digit in the “No Intervention” MLM accuracy, typing 0.892 in lieu of 0.928.